Printed Pages:02 **Sub Code:REE-101**

Paper Id: 233112 **Roll No.** | | | | | | | | | | |

B.TECH
(SEM I) THEORY EXAMINATION 2022-23
BASIC ELECTRICAL ENGINEERING

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

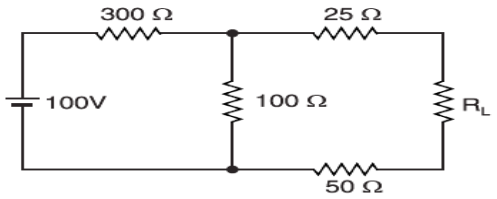
1. Attempt *all* questions in brief. **7 x 2 = 14**

(a) For ideal voltage & current source what will be the value of internal resistance.

(b) Explain source transformation by suitable example.

(c) An Alternating current is i=14.14 sin314t find its average,rms value.

(d) Find out R_L to obtain max power.



(e) Give two comparison of moving coil, dynamometer type instruments.

(f) Why Iron loss also known as constant loss.

(g) Give two comparisons between squirrel cage induction motor &slip-ring induction motor.

SECTION B

2. Attempt any *three* of the following: **3 x 7 = 21**

(a) Explain Delta to Star and Star to Delta Conversion.

(b) What are reasons / causes for low power factor, explain any one method to improve the power factor.

(c) Write advantages of 3 phase starconnected systems, A 3-phase star connected load consists of three similar inductive coils of resistance of 20 Ω and inductive reactance of 15Ω. The supply is 400 V, 50 Hz Calculate (i) the line current (ii) the power factor (iii) total power supplied.

(d) Explain fringing of magnetic flux, magnetic leakage, and hysteresis and eddy current losses.

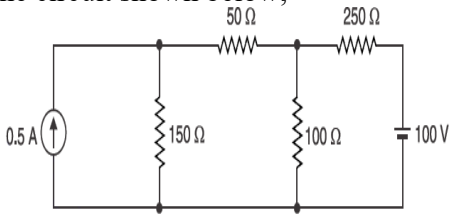
(e) Explain Double Revolving Field Theory in 1 phase induction motor; also explain shaded pole method to start the motor.

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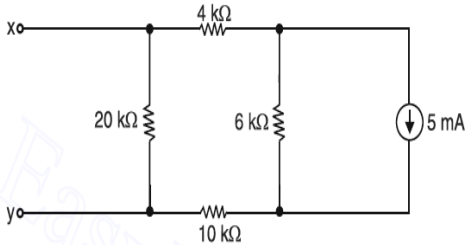
SECTION C

3. Attempt any *one* part of the following: 1 x 7 = 7

(a) Using mesh analysis, find the currents in 50 Ω, 250 Ω and 100 Ω resistors in the circuit shown below;



(b) Find the Thevenin's equivalent circuit lying to the right of terminals x – y.



4. Attempt any *one* part of the following: 1 x 7 = 7

(a) Explain form factor, peak factor; determine their value for full wave rectifier waveform.

(b) A series R-L-C circuit consists of R = 1000 Ω, L = 100 mH and C = 10 Pico farad. The applied voltage across the circuit is 100 V.
(i) Find the resonant frequency of the circuit.
(ii) Find the quality factor of the circuit at the resonant frequency.
(iii) Find upper, lower cut off frequency.
(iv) Calculate the bandwidth of the circuit

5. Attempt any *one* part of the following: 1 x 7 = 7

(a) Explain working principle of Permanent-Magnet Moving Coil (PMMC) Instruments, with advantages &disadvantages.

(b) Explain the two-wattmeter method for determination of 3 phase power and power factor of three-phase load.

6. Attempt any *one* part of the following: 1 x 7 = 7

(a) Draw BH curve; explain the eddy current losses and hysteresis losses, how these losses can be minimized.

(b) In a 250 KVA transformer has iron loss1.8 KW, full load copper loss is 2000 W. Calculate the efficiency at full load & 0.8 power factor lagging. Determine also maximum efficiency at 0.8 power factor lagging.

7. Attempt any *one* part of the following: 1 x 7 = 7

(a) Derive the expression of Generated EMF of DC Machine. Briefly explain what role Back EMF plays starting & running of motor.

(b) Draw and explain the torque- slip characteristics of a three phase induction motor, what is effect of rotor resistance on torque- slip characteristics.

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